Review 2 Work Plan: Stack Overflow in the AI Era

Oct 9, 2026 · @Rishiikesh

Overview

Review 2 (20 marks, Unit 3) is on Oct 13, 2026. The team uses two methods: **time-series analysis** and **text mining**. It adds a **Streamlit dashboard** and freezes all work by 9 pm on Oct 11, 2026. Monday is for rehearsal and fixes only.

Review 2 answers the two research questions the problem statement set aside for it:

- **RQ4, topics and complexity:** what developers still bring to human communities, and how topics, tone and question complexity changed around AI-assistant adoption (text mining).
- **RQ5, technology forecasting:** which tags keep declining, which stabilise, and how community managers should adjust (time series).

Rubric component	Marks	Lead	Support
Method 1, time series: data preparation and implementation	2	M3 Akshay (patterns), M5 Adhikkesh (forecasting)	M1
Method 1, time series: evaluation and interpretation	3	M5 Adhikkesh	M3
Method 2, text mining: data preparation and implementation	2	M2 Rishiikesh (corpus, sentiment, complexity)	M4
Method 2, text mining: evaluation and interpretation	3	M4 Nishanth (topics, keyword shift)	M2
Combined business insights and recommendations	2	M3 Akshay	everyone
Interactive dashboard	5	M1 Abhinav	each method owner supplies their page's data

Rubric component	Marks	Lead	Support
Presentation and individual contribution	3	everyone	M3 builds the deck, M1 the board evidence

Deliverables the rubric names, and where each one lives:

1. Two methods implemented and evaluated: `notebooks/07_timeseries.ipynb`, `notebooks/08_text_mining.ipynb`, code in `src/`.
2. Interactive dashboard: `dashboard/` (Streamlit), deployed with a public link.
3. Combined recommendations and final report: `reports/review2/final_report.md`, exported to PDF.
4. Final Jupyter notebook: `notebooks/review2_main.ipynb`, executed top to bottom. Plus a 15-minute team presentation.

Fixed decisions

Every member works from these decisions. Anything that changes one needs a comment on the board first.

Data (nothing new is collected).

- Volume series: `data/processed/weekly_counts.csv`. That's 14 series (`ALL` plus 13 tags) × 400 Tuesday-start weeks, 2019-01-01 to 2026-08-25, complete counts with 0 missing weeks. These are the time-series data, never the sampled records.
- Rate series: `data/processed/monthly_tag_tables.csv`, with weighted monthly resolved rate, answered rate and median hours per tag. Months flagged `low_sample` are shown greyed and never drive a conclusion.
- Text: `data/processed/questions_clean_v1.parquet`, 302,026 questions. Use `title`, `plain_body` (prose with code removed), `code_text` and full `tags` (mean 3.27 tags per question).
- Before 2025 every question carries `sampling_weight`. Any share, rate or count built from questions before 2025 is weighted. From 2025 the weight is 1.

Periods. Pre-AI = 2019-01 to 2022-10. Transition = 2022-11 to 2022-12. Post-AI = 2023-01 to 2026-08. Event markers come from `docs/event_list.csv`: Copilot GA, ChatGPT launch, the SO ChatGPT ban, the moderator strike.

Wording rule (unchanged from Review 1). Say "associated with" or "consistent with" AI adoption, never "caused by". Every forecast is reported with a prediction interval and a baseline next to it.

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One source for every number. Each method writes its results to `reports/review2/` as CSV or JSON. The notebook, dashboard, final report and slides all read those files. Nobody types a number into a slide by hand. If the CSV changes, everything changes with it.

Honesty fixes carried over from Review 1. The Review 1 time-series headline (91.7% accuracy) was one-week-ahead, tuned on the test weeks, and with no baseline. A "same as last week" forecast scores MAPE 7.62 against Holt's 8.28 on the site total. Review 2 replaces it with the evaluation in the time-series section and says so plainly if asked. Every notebook in the submission is executed by running it (`jupyter nbconvert --execute`), never by a script that writes outputs into the file.

Folder layout (new files only).

Path	Contents	Owner
`src/ts_patterns.py`	decomposition, stationarity, change points, interrupted time series	M3
`src/ts_forecast.py`	baselines, ETS, SARIMAX, rolling-origin backtest, forecasts	M5
`src/text_prep.py` , `src/sentiment.py` , `src/complexity.py`	corpus cleaning, tone, complexity metrics	M2
`src/topics.py` , `src/keywords.py`	topic models, topic trends, pre/post distinctive words	M4
`notebooks/07_timeseries.ipynb`	Method 1 walkthrough	M3 + M5
`notebooks/08_text_mining.ipynb`	Method 2 walkthrough	M2 + M4
`notebooks/review2_main.ipynb`	final combined notebook	M1

Path	Contents	Owner
`dashboard/`	Streamlit app, `dashboard/data/` small exports, its own `requirements.txt`	M1
`reports/review2/ts/`, `reports/review2/text/`	method outputs (CSV, JSON, PNG)	method owners
`reports/review2/final_report.md`	combined findings and recommendations	M3
`reports/review2/contribution_memberX.md`	individual summaries	each member

New packages (pin in `requirements.txt` on Day 1, M1): `streamlit`, `plotly`, `ruptures`, `gensim` (topic coherence only), `vaderSentiment`, `textstat`, `nbconvert`. `statsmodels`, `scikit-learn` and `spacy` are already pinned.

Git. One branch per task (`m2/text-prep`). Commits start with the issue number (`#120 add tokenizer`). Every merge goes through a pull request approved by the named reviewer.

Timeline and board rules

There are three build days (Friday to Sunday), a rehearsal day and then the review. Each day ends at a gate that the next day's work depends on. The corpus (M2) and the series file (M5) must be merged Friday night, because three other members build on them.

Three build days, a rehearsal day, then Review 2 on 13 Oct

Each box is the main deliverable of that day; a gate must pass before the next day starts

	Fri 9 Oct	Sat 10 Oct	Sun 11 Oct
M1 Abhinav Dashboard	Repo + board ready Dashboard skeleton live	Export script + test Pages wired to outputs	Deploy + fresh-clone test Final notebook executed
M2 Rishiikesh Text prep	Corpus merged (critical) corpus_stats.json	Tone + 200-label check Complexity; topic names	Tone vs resolution model Text notebook part A
M3 Akshay TS patterns	STL on 14 series Stationarity tests	Change points + ITS Tone labels (2nd rater)	Insight matrix, report Deck draft; TS notebook A
M4 Nishanth Topics	TF-IDF + 80k sampler Tested on 2,000 sample	NMF/LDA grid, pick k Final topics + names	Topic trends, keywords Text notebook part B
M5 Adhikkesh Forecasting	series_long.csv merged Baselines + backtest	ETS + SARIMAX backtest Metrics, DM, residuals	26-week forecasts, classes TS notebook part B
Gates	◆ Gate: corpus, series in	◆ Gate: all outputs merged	◆ Freeze: Sunday 9 pm
Whole team	Mon 12 Oct: two timed rehearsals, fixes only		**Tue 13 Oct: Review 2**

Review 2 schedule · 5 members × 3 build days, 3 gates

Read each row as one member's day-by-day output. If a gate slips, the owner says so on the board that night and the fallback in the risks table starts.

Board rules. The rubric warns that a board updated only just before a review is not enough evidence, so Review 2 activity must be steady across the three days.

1. By Friday 6 pm, M1 adds a "Review 2" phase. Each member creates their own issues from their table below, in Backlog, with due date, reviewer, rubric component and label set.
2. A task moves to To Do when it can start, and to In Progress with a one-line start comment.
3. Post at least one progress comment per open task per day. Every commit starts with the issue number.
4. Finish through a pull request with "Closes #N", the evidence link in the Evidence field, then Review/Testing. The named reviewer comments on what they checked, merges, and moves the task to Completed.
5. No bulk moves, no creating and closing in one sitting, and no editing another member's issue without a comment.

6. Hold check-ins at 9 am and 9 pm on Saturday and Sunday (15 minutes), noted on one "Team check-in" issue.
7. AI can write the code, but each member reads and can explain every line of their own files before merging. The rubric marks the ability to explain.

Method 1: Time-series analysis

Method 1 does two jobs. It measures the November 2022 break per tag with an interrupted time series. It then forecasts weekly volume 1, 4, 13 and 26 weeks ahead, checked against naive baselines on a rolling-origin backtest. M3 owns the patterns half (analyse time-based patterns). M5 owns the forecasting half (forecast future values).

Data preparation and implementation (2 marks)

1. **Build the series (M5, Day 1).** Pivot `weekly_counts.csv` to 400 weeks × 14 series and assert 7-day spacing with 0 gaps. Add `log1p(count)`, because volume falls from about 35,000 a week in 2020 to about 200 in August 2026, so errors must be relative. Add each tag's share of `ALL`. Build the monthly weighted resolved-rate series per tag from `monthly_tag_tables.csv`, masking `low_sample` months. Save `reports/review2/ts/series_long.csv`.
2. **Decompose (M3).** Run STL on log counts (period 52, robust) for all 14 series. Report trend, seasonal and remainder, plus seasonal strength `max(0, 1 - Var(R) / Var(S + R))`. Name the yearly dips you see, for example the late-December holiday week.
3. **Stationarity (M3).** Run ADF and KPSS on the log level and on the first difference. The result justifies differencing once (d = 1) in the models.
4. **Change points (M3).** Run `ruptures` PELT on each log series with the penalty fixed before looking at the result. List the detected break dates next to 2022-11-30 and the other events in `docs/event_list.csv`.
5. **Interrupted time series (M3).** Fit one segmented regression per series with Newey–West (HAC, lag 4) standard errors:

```
\log(y_t + 1) = \beta_0 + \beta_1 t + \beta_2\,\text{post}_t + \beta_3 (t - t_0)\,\text{post}_t + \sum_{k=1}^{2}\left[a_k \sin\tfrac{2\pi k t}{52.18} + b_k \cos\tfrac{2\pi k t}{52.18}\right] + \varepsilon_t
```

Here t0 is the week of 2022-11-30. Report the level change as exp(β2) − 1, and the weekly slope change as exp(β3) − 1, both with 95% intervals. Add a placebo break at 2021-11-30. A real effect should be large at 2022-11 and small at the placebo.

6. **Forecasting models (M5).** Every model is fitted on log counts, with hyperparameters chosen on training data only.

Model	What it is	Role
Naive	next week = this week	baseline every model must beat
Seasonal naive	next week = same week last year	baseline for the yearly pattern
Drift	random walk with the average past change	baseline for a steady decline
ETS (damped)	`statsmodels` ETSModel, additive error, additive damped trend, fitted by maximum likelihood	main smoothing model; replaces the hand-written Holt loop
SARIMAX + intervention	ARIMA(p, 1, q) with p, q ⩽ 2 picked by AIC on training data, 2 Fourier pairs for yearly seasonality, step and ramp regressors at 2022-11-30	main model that knows about the break
GRU (appendix only)	Review 1 network, run on the same backtest	kept only if it beats naive; otherwise reported as not better

7. **Backtest design (M5).** Use forecast origins every 4 weeks from 2025-01-07 to the last week that still has the horizon inside the data. Use an expanding training window and refit at every origin. Horizons are 1, 4, 13 and 26 weeks. No parameter is chosen from these origins. Model choices are fixed beforehand using 2024 H2 as a validation block.
8. **Final forecasts (M5).** Refit on all 400 weeks and forecast 26 weeks (2026-09-01 to 2027-02-23) with 80% and 95% intervals, back-transformed from log. Label it "to February 2027", not "2027".

Evaluation and interpretation (3 marks)

- **Metrics per series, model and horizon:** MASE (scaled by the naive one-step error on training), sMAPE, and RMSE on the log scale. Add skill = 1 − MASE(model) / MASE(naive), and 80%/95% interval coverage. Drop the old "accuracy = 100 − MAPE" headline.
- **One summary table:** model × horizon, averaged over 14 series, with the winner per series in a second table. Add a Diebold–Mariano test of the best model against naive at 4 and 13 weeks.

- **Residual checks for the chosen model:** a Ljung–Box test and an ACF plot. Any structure left over is reported, not hidden.
- **Expected honest result:** at 1 week ahead naive is hard to beat. The models must earn their place at 13 to 26 weeks and through well-calibrated intervals. If they don't, the slide says so.
- **Tag classes:** these thresholds are fixed now, before results. Forecast change over the next 26 weeks: below −40% = steep decline, −40% to −10% = declining, above −10% = stabilising. Each tag's class sits next to its ITS slope change and its share of `ALL` since 2023 (rising share = more resilient than the site).
- **Business reading:** the half-life of each tag's volume (weeks to halve at its post-2023 slope), and which tags hold or gain share. Also the expected weekly question load by tag through February 2027, given as an index against 2025 rather than invented staffing numbers.

Outputs in `reports/review2/ts/`: `series_long.csv`, `stl_components.csv`, `stationarity.csv`, `changepoints.csv`, `its_results.csv`, `backtest_metrics.csv`, `backtest_predictions.csv`, `forecasts_26w.csv`, `tag_classes.csv`, plus figures. The dashboard reads these files directly.

Method 2: Text mining

Method 2 answers RQ4 with three measurements on 302,026 questions:

- **Topics:** NMF topics and their weighted share before and after AI.
- **Tone (the rubric's "sentiment"):** VADER plus a frustration lexicon, checked against a hand-labelled sample.
- **Complexity:** how long, how code-heavy and how readable questions are.

M2 builds the corpus, tone and complexity. M4 builds topics and the keyword shift, and leads the evaluation.

Data preparation and implementation (2 marks)

1. **Corpus (M2, Day 1, everyone else depends on it).**
   - Text = `html.unescape(title)` + `plain_body` (titles still contain codes like `'`). `code_text` is left out of topic text and used only for complexity.
   - Lowercase. Replace URLs, file paths and long numbers with placeholder tokens.

- Protect technology tokens before stripping punctuation: `c++`, `c#`, `.net`, `node.js`, `next.js`, `vue.js`, `asp.net`.
- Lemmatise with spaCy `en_core_web_sm` (parser and NER off, `nlp.pipe` in batches).
- Stopwords = English plus a written Stack Overflow boilerplate list (trying, want, using, help, thanks, please, anyone, question, issue, problem, work, need). "error" is kept on purpose.
- Save `data/processed/text_corpus_v1.parquet` with `question_id, created_at, period, tags, sampling_weight, target_resolved_7d, clean_text, n_tokens`. It is git-ignored, with a 2,000-row sample committed.
- Report docs before and after, docs empty after cleaning, and vocabulary size in `corpus_stats.json`.

2. **Tone (M2).** Run VADER compound on title + prose only. Code words would wrongly read as emotion.
   - Add a written override list that sets technical words to neutral: kill, fatal, execute, master, abort, dead, failed, exception, error.
   - Classes: compound ≥ 0.05 positive, ≤ −0.05 negative, otherwise neutral.
   - Add lexicon flags for **frustration/urgency** (urgent, asap, stuck, frustrat*, "for hours", "for days", desperate, "nothing works", "please help") and **politeness** (thanks, thank you, appreciate).
   - Output weighted monthly shares by tag.
3. **Complexity (M2).** Compute per question: prose words, code lines, code blocks, tags per question, and Flesch reading ease of the prose (`textstat`). Add flags for error text, a why/how title, a version number in the text, and two or more study tags (multi-technology). Report weighted medians and shares by month and tag, pre vs post.
4. **Topics (M4).**
   - Fit TF-IDF (unigrams + bigrams, min_df 20, max_df 0.4, at most 30,000 terms) on a stratified sample of 80,000 questions, 5,000 per half-year from 2019 H1 to 2026 H2 (fewer when a half-year has less). Equal sampling keeps post-AI topics visible.
   - Fit NMF (nndsvda start) for k = 10, 15, 20, 25, 30, 40. Fit sklearn LDA on counts for the same k as the comparison.
   - Transform all 302,026 questions to topic weights and a dominant topic.
   - BERTopic on the 2025–26 questions (about 45,000) on a Colab GPU is a stretch cross-check only.

5. **Topic trends (M4).** Weighted monthly share of each topic (topic weight × `sampling_weight`). Compare pre vs post share change with a 95% bootstrap interval (1,000 resamples of questions). Add topic × resolved rate, topic × median hours to first answer, and topic × tag.
6. **Distinctive words (M4).** Run a weighted log-odds ratio with an informative Dirichlet prior (Monroe, Colaresi and Quinn, 2008), post vs pre, giving a z-score per term. Show the top 30 rising and falling terms overall and for python, javascript and powerbi.

Evaluation and interpretation (3 marks)

- **Choosing k and the model.** For every candidate, report topic coherence (NPMI with `gensim` on a 20,000-question held-out sample), topic diversity (unique words among each topic's top 25), stability (mean best-match Jaccard of top-10 words across 3 seeds) and NMF reconstruction error, all in `topic_model_selection.csv`. The rule is fixed now: pick the highest coherence among candidates with diversity ≥ 0.7 and stability ≥ 0.6. If none qualify, take the best coherence and say so.
- **Human check of topics.** M2 and M4 each name every topic on their own, from its top 15 words and 10 linked example titles, then compare. Report the share of topics with matching names. A topic nobody can name is labelled "mixed" and kept out of the recommendations.
- **Tone validation.** Draw 200 questions (100 pre, 100 post, seed 42). M2 and M3 each label tone (negative, neutral, positive) using a one-paragraph written rule. Report Cohen's kappa between the two people. Then score VADER against the agreed labels: accuracy, macro-F1 and a confusion matrix. If VADER is weak, the frustration flags become the main tone measure and the slide says why.
- **Extrinsic check (stretch, M4 with M5).** Add topic weights, tone and complexity to the Review 1 XGBoost pipeline and compare **validation** PR-AUC (0.482 now). The test period stays untouched, as in Review 1.
- **Interpretation the slides must deliver:**
  - Which topics gained share after 2022, meaning what people still bring to humans. Which lost share, meaning what moved to AI tools.
  - Whether tone and complexity shifted, and how each topic's resolution rate compares.
  - From the Review 1 EDA, hypotheses to test, not results: environment, setup and version-upgrade problems gain share; basic syntax and how-to questions lose it; questions get longer.

Outputs in `reports/review2/text/`: `corpus_stats.json`, `topic_model_selection.csv`, `topics_top_terms.csv`, `topic_labels.csv`, `topic_examples.csv` (title and link only), `topic_monthly_share.csv`, `topic_prepost.csv`, `topic_outcomes.csv`, `keywords_prepost.csv`, `tone_validation.csv`, `tone_monthly.csv`, `complexity_monthly.csv`, `complexity_prepost.csv`.

Interactive dashboard (Streamlit)

The dashboard is the largest single component (5 of 20). It is a multi-page Streamlit app with Plotly charts, deployed from the repo with a public link, and every chart responds to shared filters. M1 owns the app shell, data layer and deployment. Each method owner hands over their page's data files and one takeaway sentence per chart.

Global sidebar (on every page, kept in `st.session_state`): tag multi-select (13 tags + ALL), date range, period (pre / transition / post / all), event markers on/off, linear/log scale. A "Download filtered data" button and an "About the data and licence" panel sit at the bottom.

Page	What it shows	Interactions	Data files	Content owner
1. Overview	4 KPI tiles (weekly questions now vs 2022, % change, resolved rate pre vs post, median hours to accept), weekly volume with event markers	all global filters	`series_long.csv`, `monthly_tag_tables.csv`	M1
2. Patterns and the 2022 break	STL trend/seasonal/remainder, detected change points, ITS fitted line with level and slope change ± 95% CI	tag picker, placebo-date toggle	`stl_components.csv`, `changepoints.csv`, `its_results.csv`	M3
3. Forecasts	history + 26-week forecast with 80/95% bands, backtest table (MASE, skill, coverage), tag-class badges	model picker (naive, ETS, SARIMAX), horizon slider 1–26, band toggle	`forecasts_26w.csv`, `backtest_*.csv`, `tag_classes.csv`	M5

Page	What it shows	Interactions	Data files	Content owner
4. Topics	topic word bars, monthly topic share with pre/post shading, pre-vs-post change with CI (sorted), topic × tag heatmap, resolved rate by topic, 10 example titles as links	topic picker, sort by change or size	`topics_top_terms.csv`, `topic_monthly_share.csv`, `topic_prepost.csv`, `topic_outcomes.csv`, `topic_examples.csv`	M4
5. Tone and complexity	tone shares over time, frustration-flag rate, tone validation confusion matrix, complexity pre vs post, rising and falling words	metric picker, tag filter	`tone_*.csv`, `complexity_*.csv`, `keywords_prepost.csv`	M2
6. Recommendations	each recommendation with its stakeholder, the evidence (one number from each method) and a link to the chart behind it	stakeholder filter	`recommendations.csv`	M3
7. Question triage what-if (stretch)	predicted chance of an accepted answer within 7 days for a question you describe	tag, title, has code, error text, account age	saved model from M5	M5 + M1

Build rules.

- `src/export_dashboard_data.py` (M1) copies and slims the method outputs into `dashboard/data/`. The total must stay under 50 MB. The 316 MB cleaned parquet never ships.
- Load with `st.cache_data`. Every page must render in under 5 seconds on Streamlit Community Cloud.
- Chart titles state the takeaway ("python questions still fall about X% a quarter"), not the topic. One caption sentence under each chart says how to read it.
- One colour per tag across all pages, the same palette as the Review 1 deck. Pre/post shading and event lines look identical everywhere.
- Show titles only, linked to the question, never bodies.

Deployment and fallback. Deploy `dashboard/app.py` from `main` on Streamlit Community Cloud by the evening of Day 3, and test it from a fresh clone on someone else's laptop. Keep a local run (`streamlit run dashboard/app.py`) and a 2-minute

screen recording ready in case the network fails during the review.

Combined insights, final report and final notebook

The 2 marks for combined insights go to recommendations that need both methods at once. Every recommendation cites at least one time-series number and one text-mining number. Where relevant it also cites a Review 1 model result.

How the combination is built (M3 leads, Day 3 morning, all five attend for 60 minutes).

1. M3 builds `reports/review2/insight_matrix.csv` with one row per tag. Columns: forecast class and 26-week change (M5), ITS level and slope change (M3), top rising and falling topics within the tag (M4), frustration share change (M2), complexity change (M2), resolved rate pre vs post (Review 1 tables).
2. The team reads the matrix and writes 5 to 6 recommendations into `reports/review2/recommendations.csv`. Columns: finding, time-series evidence, text-mining evidence, action, stakeholder (from the problem statement's matrix), KPI to track, caveat.
3. Each recommendation names one of the three stakeholders: Q&A platform product team, developer-forum managers, or DevRel teams.

Templates to fill with real numbers. These are not conclusions yet:

- **Expert routing by topic:** topics gaining share while their resolved rate falls go to an expert queue (forum managers).
- **Moderator capacity by forecast:** shift moderation hours from steep-decline tags to stabilising ones, and refresh the forecast each quarter (platform team).
- **Documentation priorities:** rising setup, version or migration topics inside a tag become DevRel's guide backlog.
- **Asker experience:** if frustration rises and new accounts still resolve less (Review 1 SHAP), add a guided question template for first posts (platform team).
- **Early warning:** weekly volume outside the forecast's 95% band triggers a review of that tag (analytics lead).

Final report (M3 writes, M1 edits, Day 3). `reports/review2/final_report.md`, exported to PDF, at most 12 pages. Sections:

1. Executive summary (5 bullets)
2. Data recap
3. Method 1

4. Method 2
5. Combined insights and recommendations
6. Dashboard guide with link and screenshots
7. Limitations
8. How to reproduce
9. Contributions and board link

Limitations to state plainly: weights before 2025; 13 tags only; acceptance is a proxy for resolved; VADER is a general-purpose lexicon; topics are English prose only; forecasts made during a steep decline are uncertain; association is not cause.

Final notebook (M1, Day 3). `notebooks/review2_main.ipynb`, headings in rubric order:

- 0 Review 1 recap (one cell)
- 1 Method 1 preparation and implementation
- 2 Method 1 evaluation and interpretation
- 3 Method 2 preparation and implementation
- 4 Method 2 evaluation and interpretation
- 5 Combined insights and recommendations
- 6 Dashboard (link, screenshots, how to run)
- 7 Limitations
- 8 Contributions and board link

Heavy steps run in the `src/` scripts and save to `reports/review2/`. The notebook loads those files, shows every table and figure, and runs top to bottom in under 10 minutes with 0 errors. It is executed with `jupyter nbconvert --execute --to notebook --inplace` and committed with outputs. As in Review 1, M1 also publishes it to the team's Kaggle notebook.

Member 1: Abhinav (dashboard, final notebook, repo and board)

M1 owns the interactive dashboard (5 marks) and the final notebook, and keeps the repo and board clean. M1 needs each method's output files by the dates in the other members' tables. Until then, pages run on the Review 1 data or placeholder rows with the same columns.

Issue	Task	Due	Evidence	Reviewer
[M1-20]	Repo ready for Review 2: commit the Review 1 leftovers on a branch; move decks out of the root into `presentations/` (large files git-ignored); remove the duplicate root `weekly_counts.csv` and root Colab notebook; pin new packages; create the `reports/review2/` folders	Oct 9, 2026	pull request	M4
[M1-21]	Board: a "Review 2" phase and labels, every member's Review 2 issues created with due dates, reviewer and rubric component filled	Oct 9, 2026	board link	M4
[M1-22]	Dashboard skeleton: multi-page app, shared sidebar filters in `st.session_state`, palette, Overview page on `weekly_counts.csv`	Oct 9, 2026	pull request + screenshot	M4
[M1-23]	`src/export_dashboard_data.py` plus a test that checks every file's columns, so a changed CSV breaks the test, not the app	Oct 10, 2026	pull request + test	M4
[M1-24]	Wire pages 2 to 6 to the real outputs as they land; owners' takeaway captions in place	Oct 11, 2026	pull request + screenshots	M4
[M1-25]	Deploy on Streamlit Community Cloud; fresh-clone test on another laptop; 2-minute screen recording as fallback	Oct 11, 2026	public link + video	M3
[M1-26]	`notebooks/review2_main.ipynb` assembled in rubric order, executed with nbconvert, 0 errors, published to Kaggle	Oct 11, 2026	notebook + Kaggle link	M2
[M1-27]	README "Review 2" section: how to rebuild each method, export dashboard data and run the app	Oct 11, 2026	pull request	M2
[M1-28]	Opening slide, dashboard demo script and timing; contribution summary `reports/review2/contribution_member1.md`; viva answers	Oct 12, 2026	slides + summary	M3

Presents about 3:30: opening (1:00), the live dashboard demo (2:00) and the closing board slide (0:30).

Member 2: Rishiikesh (text-mining preparation, tone and complexity)

M2 owns Method 2′s data preparation and implementation (2 marks): the corpus, tone and complexity. M2 is the second rater for topic names. The corpus is on the critical path, because M4 cannot start topics without it. It must be merged by Friday night.

Issue	Task	Due	Evidence	Reviewer
[M2-20]	`src/text_prep.py`: unescape titles, protect technology tokens, placeholder tokens, spaCy lemmatising, written stopword list. Writes `text_corpus_v1.parquet` (git-ignored, 2,000-row sample committed) and `corpus_stats.json`. Unit tests on 10 hand-made titles	Oct 9, 2026	pull request + stats	M4
[M2-21]	`src/sentiment.py`: VADER on title + prose with the technical-word overrides; frustration/urgency and politeness flags; weighted monthly shares by tag in `tone_monthly.csv`	Oct 10, 2026	pull request + table	M4
[M2-22]	Tone validation: 200-question sample (seed 42, 100 pre, 100 post), a one-paragraph labelling rule, independent labels from M2 and M3, Cohen′s kappa, then VADER accuracy, macro-F1 and confusion matrix in `tone_validation.csv`	Oct 10, 2026	sample sheet + results	M4
[M2-23]	`src/complexity.py`: prose words, code lines, code blocks, tags, Flesch reading ease, flags; weighted pre/post medians and shares with bootstrap 95% intervals in `complexity_monthly.csv` and `complexity_prepost.csv`	Oct 10, 2026	pull request + tables	M4
[M2-24]	Second rater for topic names: name every topic from its top words and example titles before seeing M4′s names	Oct 10, 2026	your column in `topic_labels.csv`	M4

Issue	Task	Due	Evidence	Reviewer
[M2-25]	Interpretation: weighted logistic regression of resolved-within-7-days on tone, with tag and year as controls; 5 finding bullets on tone and complexity for the report	Oct 11, 2026	notebook section + bullets	M4
[M2-26]	`notebooks/08_text_mining.ipynb` part A (corpus, tone, complexity), executed with nbconvert	Oct 11, 2026	notebook	M4
[M2-27]	Dashboard page 5: hand M1 the final files and one takeaway caption per chart	Oct 11, 2026	files in `dashboard/data/`	M1
[M2-28]	Two slides (preparation pipeline; tone and complexity results), contribution summary, viva answers	Oct 12, 2026	slides + summary	M4

Presents about 2:30: text preparation (1:00), then tone and complexity results with the validation numbers (1:30).

In the viva, be ready to explain:

- why code is kept out of tone and topics;
- why the technical-word overrides exist (VADER reads "kill the process" as negative);
- why weights are applied before 2025;
- what kappa means and what it showed.

Member 3: Akshay (time-series patterns, combined insights, report and deck)

M3 owns the "analyse time-based patterns" half of Method 1. That covers decomposition, stationarity, change points and the interrupted time series. M3 also owns the combined insights (2 marks), the final report and the 15-minute deck. M3 is the second rater for the tone sample. M3 needs `series_long.csv` from M5 on Friday. Until then, M3 works directly from `weekly_counts.csv`.

Issue	Task	Due	Evidence	Reviewer
[M3-20]	`src/ts_patterns.py`: STL on log counts for all 14 series, seasonal strength, yearly dips named; `stl_components.csv` + small-multiples figure	Oct 10, 2026	pull request + figure	M5
[M3-21]	ADF and KPSS on level and first difference for every series; `stationarity.csv` with the decision on differencing	Oct 10, 2026	table	M5
[M3-22]	Change points with `ruptures` PELT, penalty fixed in advance; dates listed next to the events in `docs/event_list.csv`; `changepoints.csv`	Oct 10, 2026	table + figure	M5
[M3-23]	Interrupted time series per series with HAC errors and Fourier seasonality; level and slope change ± 95% CI; placebo at 2021-11-30; `its_results.csv`	Oct 10, 2026	table + forest plot	M5
[M3-24]	Second rater for the 200-question tone sample, labelled without seeing M2's labels	Oct 10, 2026	your column in the sample sheet	M2
[M3-25]	`notebooks/07_timeseries.ipynb` part A (patterns) with a takeaway under every chart, executed with nbconvert	Oct 11, 2026	notebook	M5
[M3-26]	Insight matrix and the 60-minute recommendations session; `insight_matrix.csv` and `recommendations.csv` (feeds dashboard page 6)	Oct 11, 2026	files + session notes on the issue	M1

Issue	Task	Due	Evidence	Reviewer
[M3-27]	Final report `reports/review2/final_report.md` + PDF, every number read from `reports/review2/`	Oct 11, 2026	report	M1
[M3-28]	15-minute deck from the Review 1 template: slide order, speakers and timing as in the presentation section; charts exported from the method outputs; dry run	Oct 11, 2026	deck	M1
[M3-29]	Final deck after rehearsal, contribution summary, viva answers	Oct 12, 2026	deck + summary	M1

Presents about 3:30: patterns and the 2022 break (2:15), then combined recommendations (1:15).

Member 4: Nishanth (topic modelling, keyword shift, Method 2 evaluation)

M4 owns Method 2′s evaluation and interpretation (3 marks): topic models, model selection, topic trends and the pre/post keyword shift. M4 builds on the TF-IDF and SVD work from Review 1. M4 starts Friday on the vectoriser and sampling code, using the committed 2,000-row corpus sample. The full run happens once M2′s corpus lands Friday night.

Issue	Task	Due	Evidence	Reviewer
[M4-20]	`src/topics.py` part 1: stratified 80,000-question sample (5,000 per half-year), TF-IDF settings from the method section, tested on the 2,000-row sample	Oct 9, 2026	pull request	M2
[M4-21]	Candidate models: NMF and LDA for k = 10, 15, 20, 25, 30, 40; NPMI coherence, diversity, stability across 3 seeds, reconstruction error; pick by the fixed rule; `topic_model_selection.csv` + chart	Oct 10, 2026	table + chart	M2

Issue	Task	Due	Evidence	Reviewer
[M4-22]	Final model: topic weights for all 302,026 questions, top 15 terms, 10 example titles per topic (title + link only), your topic names; `topics_top_terms.csv`, `topic_examples.csv`, `topic_labels.csv`	Oct 10, 2026	files	M2
[M4-23]	Agreement between your topic names and M2's; mark unnamed topics "mixed"	Oct 10, 2026	agreement table on the issue	M2
[M4-24]	Topic trends: weighted monthly share, pre/post change with 1,000-resample bootstrap CI, topic × tag, topic × resolved rate and hours to answer; `topic_monthly_share.csv`, `topic_prepost.csv`, `topic_outcomes.csv`	Oct 11, 2026	files + charts	M2
[M4-25]	`src/keywords.py`: weighted log-odds with Dirichlet prior, post vs pre, overall and for python, javascript, powerbi; `keywords_prepost.csv`	Oct 11, 2026	file + chart	M2
[M4-26]	`notebooks/08_text_mining.ipynb` part B (topics, keywords, evaluation) with 5 interpretation bullets, executed with nbconvert	Oct 11, 2026	notebook	M2
[M4-27]	Stretch with M5: add topic weights, tone and complexity to the Review 1 XGBoost and report the validation PR-AUC change. Dashboard page 4 files and captions to M1	Oct 11, 2026	`feature_gain_text.csv` + files	M5
[M4-28]	Two slides (topic model choice and evaluation; topic shift and keywords), contribution summary, viva answers	Oct 12, 2026	slides + summary	M2

Presents about 2:45: how the topic model was chosen and checked (1:15), then what shifted (1:30).

In the viva, be ready to explain:

- why the chosen model (NMF or LDA) won under the fixed rule, and what NPMI measures;

- why equal sampling by half-year;
- what the bootstrap interval means for a topic's share change.

Member 5: Adhikkesh (forecasting and Method 1 evaluation)

M5 owns the "forecast future values" half of Method 1 and its evaluation and interpretation (3 marks). The Review 1 code (`src/timeseries.py`, `src/timeseries_deep.py`) is the starting point. Three things change: no tuning on the test weeks, baselines on every table, and notebooks executed for real.

Issue	Task	Due	Evidence	Reviewer
[M5-20]	Commit the Review 1 time-series work as it stands. Add a note in `reports/review1/` that its numbers are superseded by Review 2. Build `series_long.csv` (log counts, tag share of ALL, monthly resolved-rate series)	Oct 9, 2026	pull request + file	M3
[M5-21]	`src/ts_forecast.py` part 1: naive, seasonal naive and drift baselines, rolling-origin backtest harness (origins every 4 weeks from 2025-01-07, horizons 1/4/13/26), MASE, sMAPE, log-RMSE, interval coverage; `tests/test_ts_forecast.py` on a synthetic series	Oct 9, 2026	pull request + tests	M3
[M5-22]	ETS damped (statsmodels ETSModel) and SARIMAX with step and ramp at 2022-11-30 plus Fourier terms; orders chosen by AIC on training only; settings fixed on 2024 H2	Oct 10, 2026	pull request	M3
[M5-23]	Backtest all 14 series; `backtest_metrics.csv`, `backtest_predictions.csv`; summary model × horizon table; Diebold–Mariano at 4 and 13 weeks; Ljung–Box and ACF for the chosen model	Oct 10, 2026	tables + figures	M3

Issue	Task	Due	Evidence	Reviewer
[M5-24]	Final 26-week forecasts to 2027-02-23 with 80/95% intervals; tag classes by the fixed thresholds; half-life and load index vs 2025; `forecasts_26w.csv`, `tag_classes.csv`	Oct 11, 2026	files + fan chart	M3
[M5-25]	Optional: rerun the GRU on the same backtest. Keep it in an appendix only if it beats naive; otherwise report it as not better	Oct 11, 2026	appendix table	M3
[M5-26]	`notebooks/07_timeseries.ipynb` part B (forecasting and evaluation) executed with nbconvert. It replaces `colab_timeseries_gpu.ipynb` in the submission	Oct 11, 2026	notebook	M3
[M5-27]	Dashboard page 3 files and captions to M1. Stretch: save the Review 1 classifier for the what-if page (page 7). Help M4 with the extrinsic text-feature check	Oct 11, 2026	files + model file	M1
[M5-28]	Two slides (backtest design; forecasts and tag classes), contribution summary, viva answers	Oct 12, 2026	slides + summary	M3

Presents about 2:45: backtest design and baselines (1:15), then forecasts, intervals and tag classes (1:30).

In the viva, be ready to explain:

- what MASE is and why it beats "100 − MAPE" here;
- why the Review 1 figure changed;
- what a 95% band covering 95% of outcomes means.

15-minute presentation

The deck has 13 slides in 15:00, with every member speaking. It runs in rubric order: Method 1, Method 2, dashboard, combined recommendations.

#	Slide (title states the finding)	Speaker	Time
1	Title, team, what changed since Review 1	M1	0:30
2	Review 2 questions (RQ4, RQ5) and the two methods	M1	0:30
3	Patterns: trend and yearly seasonality per tag (STL)	M3	1:00
4	The November 2022 break: change points and ITS level/slope change, with the placebo	M3	1:15
5	Forecast setup: rolling-origin backtest, three baselines, MASE and coverage	M5	1:15
6	Forecast results: model × horizon table, 26-week fan chart, tag classes	M5	1:30
7	Text preparation: pipeline and before/after numbers	M2	1:00
8	Tone and complexity shift, with kappa and VADER validation	M2	1:30
9	Topic model choice: coherence, diversity, stability, human agreement	M4	1:15
10	What developers still bring to humans: topic share change and rising/falling words	M4	1:30
11	Live dashboard demo: filter a tag, show its forecast, its topics, its recommendation	M1	2:00
12	Combined recommendations, each with one time-series and one text-mining number	M3	1:15
13	Limitations, board link, contributions, repo and dashboard links	M1	0:30

Speaking totals: M1 3:30, M3 3:30, M4 2:45, M5 2:45, M2 2:30.

Rules.

- Every number on a slide comes from `reports/review2/` and matches the notebook and dashboard exactly.

- One main chart per slide, readable from the back of the room.
- The demo runs on the deployed link with tabs pre-opened. The local run and the recording stay on standby.
- Rehearse twice on Monday with a timer. During questions, the member who owns the area answers first.

Viva questions and model answers

Every member should be able to give these answers in their own words. The owner answers first.

Likely question	Model answer	Owner
Why these two methods?	They answer the two research questions held back for Review 2: forecasting answers which tags keep declining (RQ5), and text mining answers what people still bring to humans (RQ4). They also feed one combined set of recommendations.	M3
Why weekly complete counts, not your sampled questions, for forecasting?	Sampled days cannot support weekly totals (monthly error averages 7.1%). The complete counts have 0 missing weeks across 400 weeks.	M5, M1
Your Review 1 forecast was 91.7% accurate. Why is it different now?	That figure was one week ahead, tuned on the test weeks, with no baseline, and "same as last week" scored slightly better. Review 2 fixes all three and reports MASE against naive.	M5
How do you know a forecast is good?	It beats naive and drift on MASE across a rolling-origin backtest at several horizons, and its 80% and 95% bands cover about 80% and 95% of what happened.	M5

Likely question	Model answer	Owner
Can a model trained before 2023 forecast after the break?	It is refitted at every origin with all data up to that point, and the SARIMAX model carries the break as step and ramp terms.	M5
Did ChatGPT cause the drop?	We cannot prove cause. The level and slope change at November 2022 is large, the placebo a year earlier is small, and other events (the answer ban, the moderator strike) happened close by. So: consistent with AI adoption.	M3
Why log counts?	Average weekly volume fell about 98% from 2020 (35,504) to 2026 so far (518), so errors must be relative. Logs make a 10% miss at 30,000 and at 300 count the same.	M3, M5
How did you choose the number of topics?	A rule written before the results: the highest NPMI coherence among candidates with diversity ≥ 0.7 and stability ≥ 0.6 across seeds. Then two people named the topics independently.	M4
Stack Overflow questions are not reviews. What does sentiment mean here?	The asker's tone, meaning frustration, urgency and politeness, measured on prose only. VADER's technical-word problems are fixed by an override list, and it is checked against 200 hand-labelled questions with kappa.	M2
Why apply sampling weights to topic and tone shares?	Before 2025 each question stands for a different number of real questions. Unweighted shares would over-represent sampled weekend days.	M2
How do the two methods combine?	Each recommendation uses a forecast or break number for a tag and a topic, tone or complexity number for the same tag. The insight matrix holds both.	M3

Likely question	Model answer	Owner
What does the dashboard add over the slides?	Any tag, period, model and horizon can be explored live from the same files as the notebook, with forecasts, topics and recommendations linked by tag.	M1
How is privacy and licence handled in the dashboard?	Titles only, linked to the original question, with a CC BY-SA 4.0 notice. No bodies, no user hashes, aggregates only.	M1, M2
Walk us through your own code.	Each member explains every function in their own scripts and notebook section, and why each choice was made.	everyone

Full-marks checklist and risks

Tick each line on Sunday night before the freeze. Anything unticked on Monday morning is cut or reported as not done, never patched in a hurry.

Method 1, time series (5)

- [ ] Series built from complete weekly counts, with the log transform and gap check shown
- [ ] STL, stationarity, change points and ITS with placebo, each with a written takeaway
- [ ] Three baselines and two models on the same rolling-origin backtest; nothing tuned on test origins
- [ ] MASE, skill vs naive and interval coverage by horizon; residual checks shown
- [ ] 26-week forecasts with 80/95% bands, labelled "to February 2027"; tag classes from the fixed thresholds

Method 2, text mining (5)

- [ ] Corpus before/after numbers, stopword and override lists written down, weights applied before 2025
- [ ] Topic model chosen by the written rule; coherence, diversity and stability table shown; human naming agreement reported
- [ ] Topic share change pre vs post with bootstrap intervals; rising/falling words

- [ ] Tone validated on 200 hand-labelled questions (kappa, macro-F1); complexity shift with intervals
- [ ] Five plain-language interpretation bullets per method

Combined insights (2)

- [ ] 5 to 6 recommendations, each with a time-series number, a text-mining number, a stakeholder, a KPI and a caveat

Dashboard (5)

- [ ] Deployed link works from a phone and a fresh browser; every chart responds to the shared filters
- [ ] Forecast, topic and recommendation pages link by tag; takeaway titles and captions everywhere
- [ ] Loads in under 5 seconds per page; licence notice; recording ready

Presentation and contribution (3)

- [ ] Deck at 15:00 ± 0:30 after two timed rehearsals; every number matches `reports/review2/`
- [ ] Five contribution summaries (Review 1 template) linking issues, pull requests and outputs
- [ ] Board: every Review 2 issue has an owner, due date, reviewer and evidence link, and moved through all five stages with comments

Submission set: `review2_main.ipynb` executed, final report PDF, dashboard link, board link, repo link, Kaggle link, contribution summaries.

Risk	Effect	Fallback	Owner
Corpus late on Friday	topics, tone and complexity all slip	M4 develops on the 2,000-row sample; spaCy swapped for a regex tokenizer if lemmatising takes over 30 minutes	M2
Topic models too slow on a laptop	no model choice by Saturday	fit on the 80,000 sample only; run the grid on Colab	M4
SARIMAX fails to converge on small tags	gaps in the backtest table	keep ETS and the baselines for those tags and say which ones	M5

Risk	Effect	Fallback	Owner
Streamlit Cloud deploy problems	no live demo	local run plus the recorded walkthrough	M1
Board looks like a last-minute burst	contribution marks at risk	issues created Friday, comments at each stage, commits spread over three days, pull requests reviewed by the named partner	everyone
Numbers drift between notebook, dashboard and slides	lost credibility in the viva	all three read `reports/review2/`; final export after the freeze only	M1, M3